

DNS Lookup Tool - Project Summary

The DNS Lookup Tool is a Python-based cybersecurity and networking utility designed to retrieve important DNS records of a target domain. The tool helps users understand how domain name systems work and assists in basic domain intelligence gathering and reconnaissance activities.

The project uses the dnspython library to perform DNS queries and fetch records such as:

- A Records
- MX Records
- NS Records

The tool provides a simple command-line interface where users can enter a domain name and receive DNS information in a clean and readable format.

Features

- Fetch A records
- Retrieve MX records
- Get NS records
- Beginner-friendly implementation
- Command-line based interface
- Improved exception handling
- ASCII banner terminal UI

DNS Record Explanation

A Records (Address Records)

A Records are used to map a domain name to its IPv4 address.

When a user visits a website, DNS uses the A Record to locate the server hosting the website.

Example:

google.com → 142.250.193.14

Purpose:

- Connect domain names to servers
- Enable website access through browsers

MX Records (Mail Exchange Records)

MX Records specify the mail servers responsible for handling email services for a domain.

Example:

gmail.com → smtp.google.com

Purpose:

- Route incoming emails
- Identify mail servers
- Support email communication systems

MX records also contain priority values:

10 mail1.example.com

20 mail2.example.com

Lower numbers indicate higher priority.

NS Records (Name Server Records)

NS Records identify the authoritative DNS servers responsible for managing a domain's DNS information.

Example:

cloudflare.com → ns1.cloudflare.com

Purpose:

- Direct DNS queries to the correct server
- Manage domain DNS configurations
- Support proper DNS resolution

Technologies Used

- Python
- Dnspython

Learning Outcomes

This project helps understand:

- DNS fundamentals
- Domain Name System architecture
- DNS record enumeration

- Python automation
- Basic reconnaissance techniques
- Exception handling in Python

Possible Future Enhancements

The project can be enhanced by adding:

- TXT record lookup
- CNAME record lookup
- Reverse DNS lookup
- Whois integration
- Subdomain discovery
- Export results to TXT/JSON
- Multi-threaded DNS enumeration
- IP geolocation lookup

⚠ Disclaimer

This project is intended for educational and authorized security testing purposes only.

🌐 30 Days of Cybersecurity Automation

Building practical cybersecurity tools, one day at a time.

Output:

```

DNS Lookup
DNS Lookup Tool

Enter Domain: cloudflare.com

[ A Records ]
[+] 104.16.133.229
[+] 104.16.132.229

[ MX Records ]
[+] 10 mxb.global.inbound.cf-emailsecurity.net.
[+] 5 mxa-canary.global.inbound.cf-emailsecurity.net.
[+] 5 mxb-canary.global.inbound.cf-emailsecurity.net.
[+] 10 mxa.global.inbound.cf-emailsecurity.net.

[ NS Records ]
[+] ns4.cloudflare.com.
[+] ns5.cloudflare.com.
[+] ns6.cloudflare.com.
[+] ns7.cloudflare.com.
[+] ns3.cloudflare.com.
```

DNS Lookup

DNS Lookup Tool

Enter Domain: github.com

[A Records]

[+] 20.207.73.82

[MX Records]

[+] 0 github-com.mail.protection.outlook.com.

[NS Records]

[+] ns-1283.awsdns-32.org.

[+] ns-1707.awsdns-21.co.uk.

[+] ns-421.awsdns-52.com.

[+] ns-520.awsdns-01.net.

[+] dns1.p08.nsone.net.

[+] dns2.p08.nsone.net.

[+] dns3.p08.nsone.net.

[+] dns4.p08.nsone.net.

DNS Lookup Completed Successfully